SQL	`SELECT * FROM people WHERE name>="Bruce" AND birthdate>"1950" AND birthdate<"1980" AND salary>=1500 AND salary<=10000`
k-d tree[21]	`tree.pointsInRectangle({name:{min:"Bruce", max: inf}, birthdate:{min:"1950", max:"1980"}, salary:{min: 1500, max:10000}})`
Operation	**Partial Search**
SQL	`SELECT * FROM people WHERE birthdate>"1950" AND birthdate<"1980" AND salary>=1500 AND salary<=10000`
k-d tree	`tree.partialRangeQuery({birthdate:{min:"1950", max:"1980"}, salary:{min: 1500, max:10000}})`
k-d tree[22]	`tree.pointsInRectangle({name:{min:-inf, max: inf}, birthdate: {min:"1950", max:"1980"}, salary:{min: 1500, max:10000}})`

11.4.4 Clustering

Finally, we get to one of the most important applications of nearest neighbor search: clustering. This application is so important that we will pledge a whole chapter, next chapter, to explaining two clustering algorithms that use NN search at their code, DBSCAN and OPTICS.

We'll provide a proper description of what clustering is in the next chapter, for now it will suffice to say that clustering is an unsupervised learning method, where a machine learning model is fed with a set of unlabeled points, and it outputs a possible grouping of these points in meaningful categories. For instance, we could develop a clustering algorithm that, given a dataset of people (with age, education, financial situation etc…) groups them in categories that share similar interests. The algorithm won't be able to tell us what these categories are, though: it's a data scientist's job, then, to study the algorithm output and see, for instance, that one category matches middle-class teenagers, another seems to fit college-graduated baby boomers, and so on… This kind of algorithm is often used to target online advertisement.